

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-20260902-59RF4X
Timestamp	2026-09-02 11:42:03
Intent	single_image_vqa
Tools Used	vqa, grounding, geospatial

Query

Describe the scene and report land-cover features.

Final Answer

Scene analysis complete.

=== Local CV VQA Prototype ===

Scene contains dense built-up structures, road networks or linear structures.

Illumination appears normal.

Approximately 374 distinct structural regions detected.

Evidence Score: 75/100.

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.884	0.30
Sensor Reliability	S	0.840	0.20
Spatial Consistency	P	0.500	0.20
Temporal Consistency	T	0.820	0.15
Input Quality	I	0.616	0.15
FINAL SCORE		74.9 / 100	

Validation

Status: **PASS** ✓

- 1 image(s) detected — count OK.
- Image 1: Quality OK (mean=112, std=43.0).

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km ²)	Demo Georeferenced
1	candidate regions of interest	0.97	249.345	Yes

Specialist Tool Outputs

VQA — SUCCESS — 3 region(s)

=== Local CV VQA Prototype ===
Scene contains dense built-up structures, road networks or linear structures.
Illumination appears normal.
Approximately 374 distinct structural regions detected.

GROUNDING — SUCCESS — 1 region(s)

Grounding found 1 candidate region(s) matching 'regions of interest'.
Highest confidence: 0.97.

GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 249.34 km².

■ Approximate demo georeferencing applied.
Demo center: (20.296°N, 85.824°E)
Coordinates are NOT derived from real GeoTIFF metadata.

Execution Trace

[11:42:03] QUERY RECEIVED: Describe the scene and report land-cover features.
[11:42:03] ROUTER DECISION: Intent: single_image_vqa
Selected tools: vqa, grounding, geospatial
Images required: 1
[11:42:03] VALIDATION: Status: PASS
1 image(s) detected – count OK.
Image 1: Quality OK (mean=112, std=43.0).
[11:42:03] TOOL EXECUTING: vqa → started
[11:42:04] TOOL OUTPUT (vqa): Success – 3 region(s) detected
=== Local CV VQA Prototype ===
Scene contains dense built-up structures, road networks or linear structures.
Illumination appears normal.
[11:42:04] TOOL EXECUTING: grounding → started
[11:42:04] TOOL OUTPUT (grounding): Success – 1 region(s) detected
Grounding found 1 candidate region(s) matching 'regions of interest'.
Highest confidence: 0.97.
[11:42:04] TOOL EXECUTING: geospatial → started
[11:42:04] TOOL OUTPUT (geospatial): Success – 0 region(s) detected
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 249.34 km².
[11:42:04] EVIDENCE FUSION: Score: 74.9/100
M=0.884
S=0.840
P=0.500
[11:42:04] FINAL SCORE: 75/100
[11:42:04] ANSWER: Scene analysis complete.

=== Local CV VQA Prototype ===
Scene contains dense built-up structures, road networks or line

Tool Versions

Component	Version
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SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge